

"Osprey 3.5 V6: P0087 low fuel rail pressure from worn high-pressure fuel pump"

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Area: bulletins

1 Condition

Owners of Osprey 3.5 V6 vehicles may report a long crank, hesitation under acceleration, and a stored DTC P0087 (Fuel Rail/System Pressure - Too Low) with an illuminated MIL. Symptoms typically worsen under high fuel demand such as hard acceleration or grade climbing, where commanded rail pressure cannot be met. Confirm the complaint by monitoring desired versus actual fuel rail pressure with a scan tool before condemning any component. Background on how this fault is set is provided in Osprey Fuel Service Manual :: Fuel Pressure and Sensor Faults .

2 Cause

The condition is caused by accelerated wear of the high-pressure fuel pump cam follower and plunger assembly, part number HPFP-4120, which reduces the pump's ability to maintain commanded rail pressure. In a smaller subset of cases the fuel rail pressure sensor FRS-4180 reports a biased-low value that mimics true low pressure, so both components must be evaluated. Refer to Osprey Fuel Service Manual :: High-Pressure Fuel Pump (GDI) for operating principles, and to Osprey Fuel Service Manual :: Fuel Delivery for the overall delivery architecture.

3 Repair Procedure

Replace the high-pressure fuel pump HPFP-4120 following PROC-FUEL-PUMP, observing all fuel-pressure relief and cleanliness precautions for the GDI system. Detailed removal, torque, and priming steps are documented in Osprey Fuel Service Manual :: Fuel Pump Replacement Procedure . If diagnosis points to the sensor rather than the pump, replace FRS-4180 per Osprey Fuel Service Manual :: Fuel Rail Pressure Sensor Replacement . After repair, clear P0087 and confirm actual rail pressure tracks the commanded value across the operating range.

4 Parts Information

This repair uses high-pressure fuel pump HPFP-4120 and, where indicated by diagnosis, fuel rail pressure sensor FRS-4180. Both parts are specific to the Osprey 3.5 V6 GDI fuel system and must not be carried over from prior-pressure variants. Confirm part applicability and any required seals or O-rings against the Osprey Fuel Service Manual :: Fuel Rail and Pressure Sensing section before ordering. For general system diagnosis prior to parts ordering, see Osprey Fuel Service Manual :: Fuel System Diagnosis .